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22CS912 CLOUD SECURITY FOUNDATIONS

Batch / Year / Semester: 2023-2027 / IV / VII

Date: 09-05-2026

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1. Course Objectives

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3	0	0	3

OBJECTIVES:

The Course will enable learners to:

- Learn the basics of cloud security, including the shared responsibility model and identity management.
- Set up a secure cloud infrastructure with features like virtual private clouds and security groups.
- Develop skills for identifying and managing security incidents in the cloud, adhering to best practices.
- Safeguard application data at rest and in transit using encryption.
- Understand the features of Database Security and Security in Data Warehouses.

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2. Syllabus

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L	T	P	C
3	0	0	3

UNIT I	INTRODUCTION TO SECURITY IN CLOUD	09
Introduction to Security, Security in the Cloud, Security design principles, Shared responsibility model, Activity: Shared Responsibility Model, Identity and Access Management (IAM) fundamentals, Authenticating and Authorizing with IAM, Examples of authorizing with IAM, Additional authentication and access management services, Using Organizations.		
UNIT II	**SECURING INFRASTRUCTURE**	**09**
Structure of a three-tier web application, virtual private cloud (VPC), Setting up public and private subnets and internet protocols, Security groups, Network access control lists (ACLs), Load balancers, Protecting compute resources.		
UNIT III	**PROTECTING APPLICATION DATA**	**09**
Basics on Data Protection, Protect data at rest, Amazon S3 protection features, Protection through encryption, Protect data in transit, protect data in Amazon S3, additional data protection services.		
UNIT IV	**LOGGING AND MONITORING**	09
Importance of logging and monitoring, Capture and collect, Reading a Log File, AWS services with built-in logs, Monitor and report, CloudTrail and Amazon CloudWatch, methods for logging and monitoring, additional AWS services for logging and monitoring, AWS Security Hub.		
UNIT V	**RESPONDING AND MANAGING AN INCIDENT**	09
Identifying an incident, Services that support the discovery and recognition phase, AWS Config and AWS Lambda, Services that support the resolution and recovery phase, Best practices for handling an incident.		

TOTAL: 45 PERIODS

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TEXT BOOKS:

1. Tim Mather, Subra Kumaraswamy, Shahed Latif, "Cloud Security and Privacy, An Enterprise Perspective on Risks and Compliance", Oreilly Media 2009.

2. Vic (J.R.) Winkler, "Securing the Cloud, Cloud Computer Security Techniques and Tactics", Syngress, April 2011.

REFERENCES:

1. Rajkumar Buyya, James Broberg, Andrzej, —Cloud Computing: Principles and Paradigms‖, Wiley India Publications 2011.

2. Arshdeep Bahga and Vijay Madisetti, —Cloud Computing –A Hands on Approach‖, Universities Press (India) Pvt Ltd. 2014

3. Pre-requisites

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4. Course Outcomes

At the end of this course, the students will be able to:

Course Outcomes		HKL
CO1	Learn essential cloud security principles and identity management fundamentals	K3
CO2	Design and implement secure cloud infrastructure components for effective resource protection.	K3
CO3	Explore methods for securing application data stored in the cloud, including encryption techniques and data protection features	K4
CO4	Gain proficiency in implementing logging and monitoring practices to detect and respond to security events efficiently in cloud environments	K4
CO5	Acquire skills to identify and manage security incidents in the cloud, utilizing appropriate tools and techniques for incident detection, analysis, and resolution	K6
CO6	Proficiently ensure the protection of cloud environments by applying strong security measures across all aspects, ensuring resilience and compliance	K6

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5. CO-PO Mapping

Course Outcomes	Program Outcomes											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3			2	-	-	-	-	-	-	-
CO2	2	2	3		3	-	-	-	-	-	-	-
CO3	2	2		2	2	-	-	-	-	-	-	-
CO4	2	2		3	3	-	-	-	-	-	-	-
CO5		3		3	3	-	-	-	-	-	-	-
CO6	3	3	3	3	3	-	-	-	-	-	-	-

6. Lecture Plan: Unit I

Sl.No.	Topic	No. of. Periods	Proposed Date	Actual Date	Pertaining CO	Taxonomy Level	Mode of Delivery
1	Importance of logging and monitoring	1			CO4	K1	MD 1, MD 2
2	Capture and collect	1			CO4	K1	MD 1, MD 2
3	Reading a Log File	1			CO4	K1	MD 1, MD 2
4	AWS services with built-in logs	1			CO4	K1	MD 1, MD 2
5	Monitor and report	1			CO4	K1	MD 1, MD 2
6	CloudTrail and Amazon CloudWatch						

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7	Methods for logging and monitoring	1			CO4	K1	MD 1, MD 2
8	additional AWS services for logging and monitoring	1			CO4	K1	MD 1, MD 2
9	AWS Security Hub	1			CO4	K1	MD 1, MD 2

ASSESSMENT COMPONENTS	MODE OF DELEIVERY
AC 1. Unit Test AC 2. Assignment AC 3. Course Seminar AC 4. Course Quiz AC 5. Case Study AC 6. Record Work AC 7. Lab / Mini Project AC 8. Lab Model Exam AC 9. Project Review	MD 1. Oral presentation MD 2. Tutorial MD 3. Seminar MD 4. Hands On MD 5. Videos MD 6. Field Visit

7. Activity Based Learning

Learning Method	Activity
Learn by Questioning	Quiz / MCQ Using RMK Nextgen App

RMK Nextgen

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8. Lecture Notes

UNIT IV LOGGING AND MONITORING

Importance of logging and monitoring, Capture and collect, Reading a Log File, AWS services with built-in logs, Monitor and report, CloudTrail and Amazon CloudWatch, methods for logging and monitoring, additional AWS services for logging and monitoring, AWS Security Hub.

Importance of logging and monitoring

Logging is the collection and recording of activity and event data. The service itself can provide logging capabilities, as with Amazon Virtual Private Cloud (Amazon VPC) Flow Logs and Amazon Simple Storage Service (Amazon S3) server access logs. Or a secondary service, such as AWS CloudTrail, might provide logging.

The type of information logged will vary based on the service that is conducting the logging, but some common elements are logged:

- Date and time of event
- Originating location of request
- Identity of resources accessed

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What is logging?

- Logging is the collection and recording of activity and event data.
  - Provided by the service itself
  - Provided by a secondary service
- Information logged will vary based on the service conducting the logging.
- Common log elements:
  - Date and time of event
  - Origin of event
  - Identity of resources accessed


Mar 31 23:17:22] 127.0.0.1
list.php HTTP/1.1" 200 189
[Mar 31 23:17:22] 127.0.0.1
onnect.php?__user=
sg=AQDJ95ij HTTP/1.1" 200 561
[Mar 31 23:17:22] 127.0.0.1
_list.php HTTP/1.1" 200 189
[Mar 31 23:17:22] 127.0.0.1
.swf?v=1 HTTP/1.1" 200 21115
[Mar 31 23:18:32] 127.0.0.1
partition=176&cb=12s6 HTTP/1.1"
[Mar 31 23:18:32] 127.0.0.1
[Mar 31 23:18:32] 127.0.0.1
- [Mar 31 23:18:32] 127.0.0.1
- [Mar 31 23:18:32] 127.0.0.1
- [

A comprehensive logging methodology can help you in every phase of your incident response strategy. Log files are also a primary focus point in incident response. This is because of their ability to provide detailed, time-stamped records to investigators and incident responders. Logging provides a record of events captured at a specific point in time. Common log files include access logs, application logs, system logs, event logs, API call logs, and database logs. Log files can assist in troubleshooting performance issues within your AWS Cloud and on-premises environment. Logs are also necessary to perform security audits and adhere to recordkeeping requirements to demonstrate compliance with a number of regulatory measures. Examples of regulatory measures include the Health Insurance Portability and Accountability Act (HIPAA) in the United States, the European Union General Data Protection Regulation (GDPR), and Brazil's General Data Protection Law (LGPD). Finally, log files can assist you to remediate issues that arise organically or through audits. Logs provide you with insights on the point in time when an issue arose.

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What is monitoring?

- Monitoring is the continuous verification of the security and performance of your resources, applications, and data.
- AWS provides several services that give you the visibility to spot issues before they impact operations:
  - AWS CloudTrail
  - Amazon CloudWatch
  - Amazon EventBridge
  - AWS X-Ray



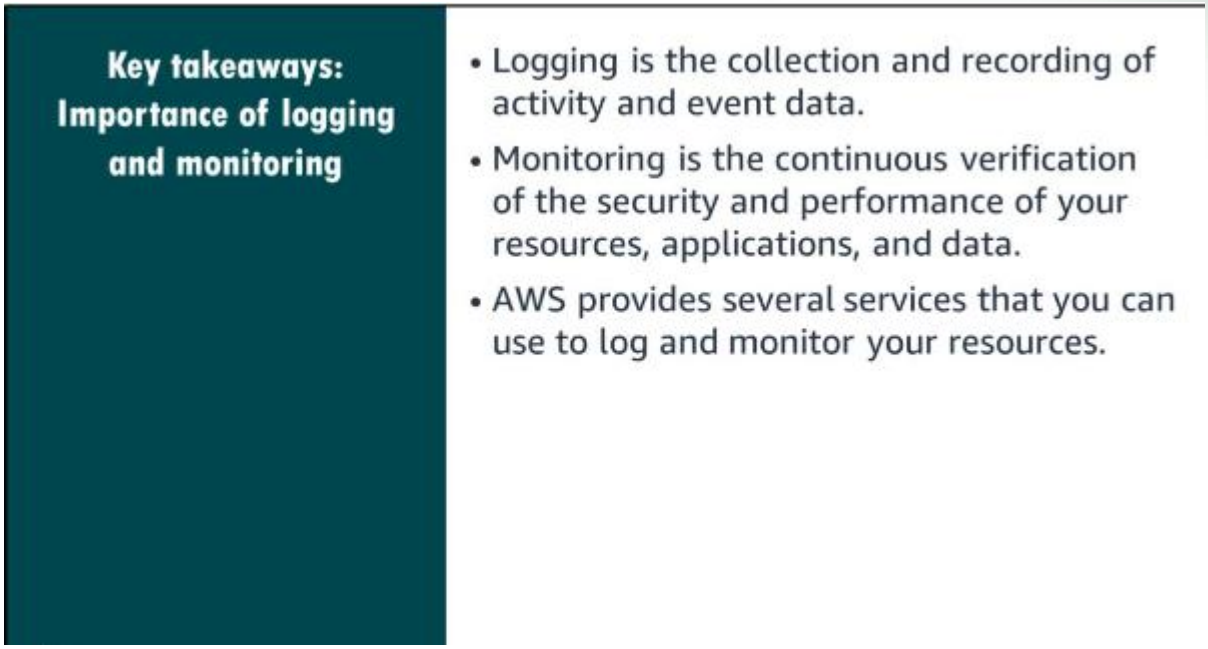
Monitoring is the practice of continuously verifying the security and performance of your resources and data. AWS provides several services that you can use to monitor your resources, but this module will focus on two: **AWS CloudTrail** and **Amazon CloudWatch**.

CloudTrail provides a record of actions taken within your environment. CloudTrail logs include information such as the action type, identity of the user, and time and date of the action. With this information, you can monitor who is doing what and when they are doing it.

With CloudWatch, you can monitor your resources and applications in real time. CloudWatch provides you with system-wide visibility into resource utilization, application performance, and operational health.

Monitoring services and tools, along with a solid incident response plan, can help you mitigate the effects of a system outage or malicious actor. In many cases, monitoring can help you to spot an issue before it has operational impact.

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Key takeaways from this section of the module include the following:

- Logging is the collection and recording of activity and event data.
- Monitoring is the continuous verification of the security and performance of your resources, applications, and data.
- AWS provides several services that you can use to log and monitor your resources.

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Capture and collect

AWS CloudTrail

- Assists you to enable governance and compliance, as well as operational and risk auditing of your AWS account
- Records actions taken by a user, role, or AWS service as events
- Provides visibility of events in the CloudTrail console
- Can be used to view, search, download, archive, analyze, and respond to account activity across your AWS infrastructure



AWS CloudTrail

AWS CloudTrail is an AWS service that helps you enable governance and compliance, as well as operational and

risk auditing of your AWS account. Actions taken by a user, role, or an AWS service are recorded as events in

CloudTrail. Events include actions taken in the AWS Management Console, AWS Command Line Interface (AWS

CLI), and AWS Software Development Kits (SDKs) and APIs. You can view, search, download, archive, analyze,

and respond to these recorded events in the CloudTrail console.

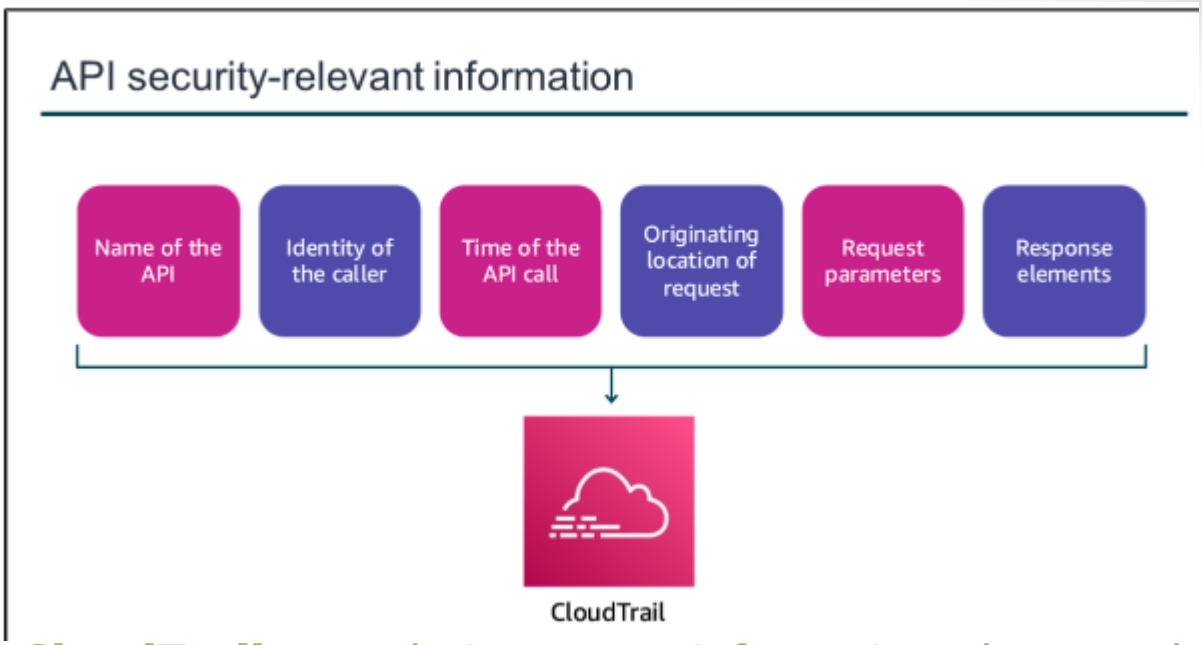
You can integrate CloudTrail into applications by using the API, automate trail creation for your organization,

check the status of trails you create, and control how users view CloudTrail events.

For more information, see AWS CloudTrail at

https://aws.amazon.com/cloudtrail

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CloudTrail records important information about each API call. Information includes the name of the API, identity of the caller, time of the API call (captured in Universal Time Coordinated, or UTC), location that the call originated from, request parameters, and response elements returned by the AWS service. This information helps you to track changes made to your AWS resources, troubleshoot operational issues, and ensure compliance with internal policies and regulatory standards.

You can use the AWS API call history that CloudTrail produces to track changes to AWS resources. Changes include creation, modification, and deletion of AWS resources, such as Amazon Elastic Compute Cloud (Amazon EC2) instances and Amazon VPC security groups.

The log file record example on the next few slides shows actions from an AWS Identity and Access Management (IAM) user named Jane. Jane used the ec2-stop-instances command in the AWS CLI to call the Amazon EC2 **StopInstances** action.

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Activity: Reading a Log File



Reading a log: Identity of the caller


```
{
  "Records": [{
    "eventVersion": "1.0",
    "userIdentity": {
      "type": "IAMUser",
      "principalId": "AIDACKCEVSQ6C2EXAMPLE",
      "arn": "arn:aws:iam::111122223333:user/Jane",
      "accountId": "111122223333",
      "accessKeyId": "AKIAIOSFODNN7EXAMPLE",
      "userName": "Jane"
    },
```


This log file was generated when someone or something performed some kind of action. Take a moment to review this log file snippet, and then answer the following questions individually or as a group:

1. What type of account did the log collect information about?

2. What can you determine from the **arn** element? On the next slide, you will be able to see when this call was made.

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Reading a log: Time and origin of the request


```
"eventTime": "2021-07-06T21:01:59Z",
"eventSource": "ec2.amazonaws.com",
"eventName": "StopInstances",
"awsRegion": "us-east-2",
"sourceIPAddress": "203.0.113.176",
"userAgent": "ec2-api-tools 1.6.12.2",
```


Take a moment to review this log file snippet, and then answer the following questions individually or as a group:

1. What does the **eventSource** field give you information about?

2. In the **eventName** field, what does the **StopInstances** value indicate?

3. What method was used to perform this action? (Console, AWS CLI, or other)

Now let's see what was involved in this request.

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Reading a log: Request parameters and response elements


```
"requestParameters": {
  "instancesSet": {
    "items": [{
      "instanceId": "i-ebeaf9e2" } ] },
  "force": false },
"responseElements": {
  "instancesSet": {
    "items": [{
      "instanceId": "i-ebeaf9e2",
      "currentState": {
        "code": 64,
        "name": "stopping" },
      "previousState": {
        "code": 16,
        "name": "running" } } ] },
```

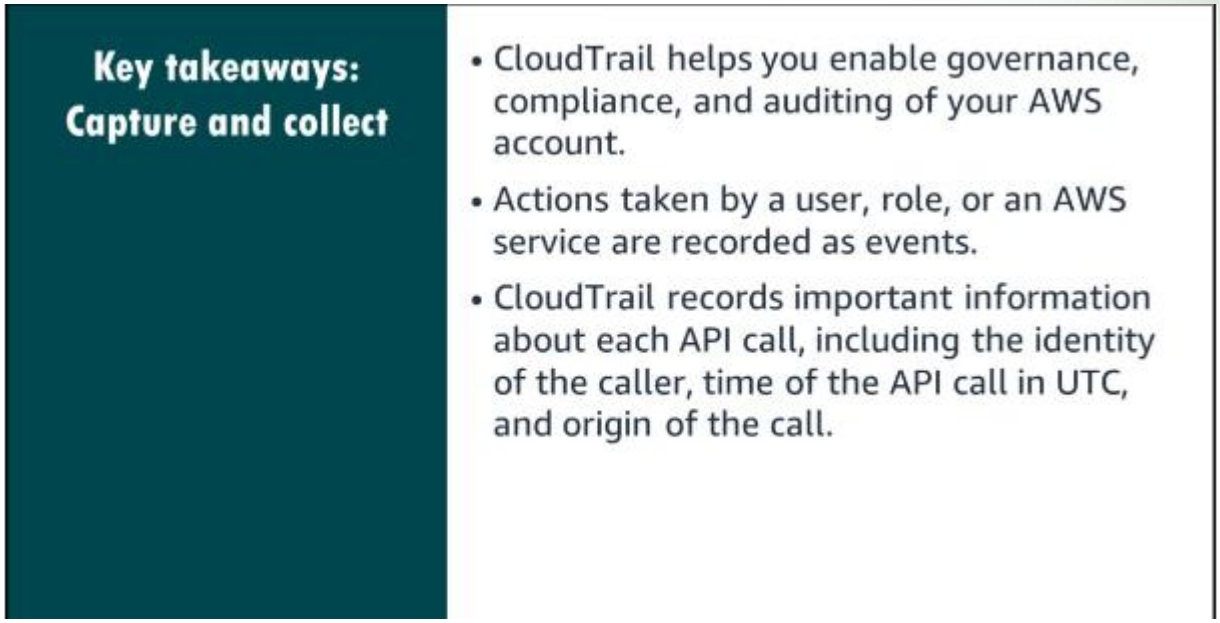

Take a moment to review this log file snippet, and then answer the following questions individually or as a group:

1. In the **instanceId** field, what does the **i-ebeaf9e2** value indicate?

2. What was the action that was performed?

If you need to track changes to such resources, answer questions about user activity, demonstrate compliance, troubleshoot, or perform security analysis, you can use **CloudTrail** to detect threats and provide security.

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Key takeaways from this section of the module include the following:

- **CloudTrail** helps you enable governance, compliance, and auditing of your AWS account.
- Actions taken by a user, role, or an AWS service are recorded as **events**.
- CloudTrail records important information about each **API call**, including the **identity of the caller**, **time of the API call in UTC**, and **origin of the call**



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Services with built-in logs: Amazon S3

- Amazon S3 provides detailed access request records through Amazon S3 server access logging.
- Server access logs provide useful information for security and access audits.
- Server access logs can provide insight into your customer base and assist you to understand your Amazon S3 bill.



Amazon Simple Storage Service (Amazon S3)

Server access logging is a built-in feature of Amazon S3. Enable the feature for detailed records of requests that are made to an S3 bucket. These logs are useful in a number of ways—from providing information during a security and access audit to providing insights about your customer base. Server access logs can also provide insights into your storage usage, which can help you to better understand your Amazon S3 bill.

For more information, see **Logging Requests Using Server Access Logging** in the Amazon S3 User Guide at:

https://docs.aws.amazon.com/AmazonS3/latest/userguide/ServerLogs.html

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Services with built-in logs: Amazon VPC

- With VPC Flow Logs, you can capture information about inbound and outbound IP traffic from the following:
  - VPC
  - Subnets
  - Individual network interfaces
- Publish flow log data to CloudWatch or Amazon S3.
- Flow log data is collected outside of the path of your network traffic, with no impact on throughput or latency.



Amazon Virtual Private Cloud (Amazon VPC)

VPC Flow Logs is a built-in feature of **Amazon VPC**. Flow logs capture information about inbound and outbound IP traffic from your VPC network interfaces. Flow logs can be captured at the VPC level, the subnet level, or for each individual network interface. You can publish flow log data to the **Amazon CloudWatch Logs** service or **Amazon S3**.

Flow logs are helpful to troubleshoot, monitor, and analyze the flow of IP traffic within your VPC. Because flow log data is collected outside of the path of your network traffic, there is no impact on throughput, latency, or overall performance.

For more information, see **Logging IP Traffic with VPC Flow Logs** in the Amazon VPC User Guide at:

https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs.html

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Services with built-in logs: ELB

- ELB access logs capture detailed information about requests sent to your load balancer.
- Use access logs to analyze traffic patterns and for troubleshooting.
- ELB captures, compresses, and stores logs in a specified S3 bucket.



Elastic Load Balancing (ELB)

Elastic Load Balancing (ELB) access logs capture detailed information about requests sent to your load balancer. Logs contain information about the time of request receipt, IP address of the client, latencies, request paths, and server responses. The information contained in these logs can assist you to troubleshoot issues and analyze traffic on your network.

For more information, see **Access Logs for Your Application Load Balancer** in the User Guide for Application Load Balancers at:

https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-access-logs.html

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Monitor and report

Logging and Monitoring

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Amazon CloudWatch

- Is a monitoring and observability service
- Provides a unified view of the operational health of your AWS resources, applications, and services
- Collects metrics in the AWS Cloud and on premises
- Can be used for infrastructure monitoring and troubleshooting
- Provides the ability to customize logs and events



Amazon CloudWatch

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Amazon CloudWatch is a monitoring and observability service from AWS designed for DevOps engineers, developers, site reliability engineers, IT managers, and product owners. It collects operational data in the form of logs, metrics, and events from AWS resources, applications, and on-premises servers, providing a unified view for performance tracking and troubleshooting. CloudWatch enables users to detect anomalous behavior, set alarms, visualize metrics and logs together, take automated actions, and gain system-wide insights to ensure smooth application performance. With CloudWatch Events (now being replaced by Amazon EventBridge), users can receive near-real-time notifications of changes in AWS resources and automatically trigger actions such as invoking Lambda functions or sending alerts. CloudWatch Logs centralizes log collection from EC2 instances, CloudTrail, Route53, and other sources, allowing storage, search, filtering, querying with CloudWatch Logs Insights, and visualization in dashboards. By consolidating data from different AWS and on-premises sources, CloudWatch helps break down data silos, improve troubleshooting, and automate responses, making it a key tool for maintaining operational efficiency across cloud environments.Ask ChatGPT

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Comparing CloudTrail and CloudWatch

AWS CloudTrail	Amazon CloudWatch
Continuously monitors and logs user activities	Continuously monitors resource and application performance
Useful for compliance auditing, security analysis, and troubleshooting	Useful for detecting anomalous service behavior, setting alarms, and discovering insights
Helps you determine WHO performed WHAT unauthorized action and WHEN they did it	Alerts you that an issue has occurred due to an unauthorized action

When used together, you can create custom CloudWatch dashboards, alarms, and notifications for key metrics and specific CloudTrail events.

The **AWS CloudTrail** service provides the ability to continuously monitor and log the activity of users, groups, and roles within your AWS environment. These log files are useful for compliance auditing, security analysis, and troubleshooting. Assume that an EC2 instance was deleted without permission. CloudTrail could provide a log that gives you the identity of the user, group, or role that performed the unauthorized action.

Amazon CloudWatch provides the ability to continuously monitor the performance of your AWS resources and applications. CloudWatch can help detect and notify you of anomalous service behavior, such as an EC2 instance deletion. Whether coupled with CloudTrail or on its own, CloudWatch provides you with increased oversight of your AWS environment.

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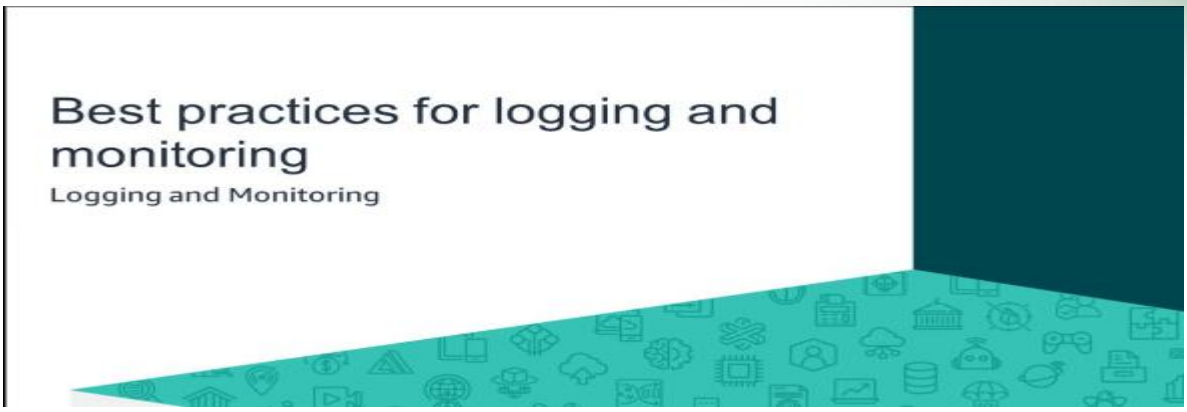
When used together, you can create custom **CloudWatch alarms** and **notifications** for specific **CloudTrail events**. This helps you to respond quickly to key operational issues.



Key takeaways from this section of the module include the following:

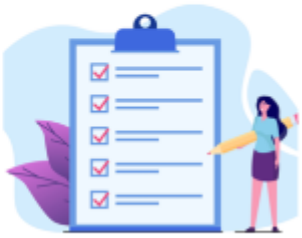
- **CloudWatch** provides a unified view of the operational health of your AWS resources, applications, and services.
- CloudWatch collects monitoring and operational data as **logs**, **metrics**, and **events**.
- **CloudTrail monitors actions**, and **CloudWatch monitors performance**.
- Create custom **dashboards**, **alarms**, and **notifications** for key metrics.

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Best practices for logging and monitoring

- Define your organizational requirements for logs, alerts, and metrics.
- Configure service and application logging throughout your workload.
- Analyze your logs centrally.



The first step to use the logging and monitoring capabilities that AWS provides is to **define your requirements**. Identify the **resources, applications, and services** that you want to maintain logs for. Logging and monitoring requirements can vary widely, so you must determine what the **organizational, legal, and compliance requirements** for your workloads are. Then, evaluate and identify the resources that AWS has available to assist.

When you **collect metrics and define baselines**, you can gain insights into potential **security threats**. Define **who should receive alerts** and **what they should do** with the alerts that they receive. Configure logging throughout the workload, including **application logs**, **AWS services logs**, and **resource logs**

Collect your logs centrally. Use automation from services such as **CloudWatch** to analyze logs to detect any **anomalies or indicators of malicious activity or compromise**.

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AWS Trusted Advisor

- Provides recommendations based on five categories of AWS best practices: cost optimization, security, fault tolerance, service limits, and performance improvement
- Evaluates your account to suggest improvements and optimizations for your resources
- Is accessible through the AWS Management Console and available to all support tiers



AWS Trusted Advisor

AWS Trusted Advisor provides recommendations that help you follow AWS best practices, which have been learned from serving hundreds of thousands of AWS customers. Trusted Advisor evaluates your account by using checks based on five categories of AWS best practices. The checks identify ways to **optimize your AWS infrastructure**, **improve security and performance**, **reduce costs**, and **monitor service quotas**.

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Assume you're the administrator of your organization's AWS account. You're looking for ways to optimize your account resources and improve your overall security posture, but the time it would take to do so manually would be prohibitive. AWS Trusted Advisor can automate this process for you, providing you with recommendations for actions you can take to improve these areas. You can then follow the recommendations to optimize your resources and security posture.

Trusted Advisor is available in all **AWS Support plans**.

- **AWS Basic Support** and **AWS Developer Support** customers can access **core security checks** and **all checks for service quotas**.

- **AWS Business Support** and **AWS Enterprise Support** customers can access **all checks**, including **cost optimization**, **security**, **fault tolerance**, **performance**, and **service quotas**.

For more information, see **AWS Trusted Advisor** in the AWS Support User Guide at: https://docs.aws.amazon.com/awssupport/latest/user/trusted-advisor.html

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Amazon EventBridge

- Is a serverless event bus service that is used to connect your applications with data from a variety of sources
- Provides a stream of real-time data from applications and services to targets, such as AWS Lambda or event buses
- Was formerly called Amazon CloudWatch Events



Amazon EventBridge

AWS Security Hub is a service that assists you to monitor your cloud security posture through the use of automated, continuous security best practice checks against your AWS resources. Security Hub aggregates security alerts from various AWS services and third-party partner products, and presents them in a standardized format, making it easier for you to act on the alerts.

You can also use Security Hub to create automated **response**, **remediation**, and **enrichment workflows** by taking advantage of Security Hub's integration with **EventBridge**. Security Hub provides a **security score** for each enabled standard and a **total score** for all accounts associated with your administrator account. This information can assist you to monitor your overall security posture.

One example of how AWS Security Hub can help is by assisting you in **prioritizing the response and remediation efforts** of your central security teams and DevSecOps teams by searching, correlating, and aggregating diverse security findings by accounts and resources.

For more information, see **AWS Security Hub** at: https://aws.amazon.com/security-hub

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AWS Config

- Helps to assess, audit, and evaluate the configurations of your AWS resources
- Continuously monitors and records AWS resource configurations
- Provides automated evaluation of recorded configurations against desired configurations



AWS Config

With the **AWS Config** service, you can **assess, audit, and evaluate** the configurations of your AWS resources. AWS Config continuously monitors and records your AWS resource configurations, which provides you with the ability to **automate the evaluation** of those configurations against desired configurations. You can also use AWS Config to **view IAM policies** that are assigned to IAM users, groups, or roles at any time that AWS Config was recording. This capability can aid you in determining what permissions belonged to what user at that specific time

AWS Config can assist you to maintain **auditing and compliance** by providing **historical resource configurations**. With the wide range of AWS Config capabilities, you can simplify **compliance auditing**, **security analysis**, **change management**, and **operational troubleshooting** in your AWS environment.

An example of an area where AWS Config is a valuable asset is **change management**. When your resources are created, updated, or deleted, AWS Config streams these configuration changes to **Amazon Simple Notification Service (SNS)**, so that you are notified of all the configuration changes. AWS Config represents **relationships between resources** so that you can assess how a change to one resource may impact other resources.For more information, see **AWS Config** at: https://aws.amazon.com/config

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Demonstration: Security Hub



Lab: Monitoring and Alerting with CloudTrail and CloudWatch



You will now complete the Monitoring and Alerting with CloudTrail and CloudWatch lab.

In this lab, you will configure logging and monitoring in an AWS account.

Lab: Tasks

1. Creating a CloudTrail trail with CloudWatch Logs enabled
2. Creating an SNS topic and subscribing to it
3. Creating an EventBridge rule to monitor security groups
4. Creating a CloudWatch alarm based on a metrics filter
5. Querying CloudTrail logs by using CloudWatch Logs Insights

In this lab, you will complete the following tasks:

1. **Creating a CloudTrail trail with CloudWatch Logs enabled**

2. **Creating an SNS topic and subscribing to it**

3. **Creating an EventBridge rule to monitor security groups**

4. **Creating a CloudWatch alarm based on a metrics filter**

5. **Querying CloudTrail logs by using CloudWatch Logs Insights**

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It's now time to review the module, and wrap up with a knowledge check and discussion of a practice certification exam question.

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9. Assignments

Question 1:

Importance of Logging and Monitoring in AWS Cloud Security

Question 2:

AWS CloudTrail and Amazon CloudWatch for Monitoring

Question 3:

Methods for Logging and Monitoring in AWS

Question 4:

AWS Services with Built-in Logs and Security Monitoring

Question 5:

AWS Security Hub for Centralized Security Management

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10. Part A Questions & Answers

1. What is the purpose of logging in cloud environments?

To record activities and events for auditing, debugging, compliance, and security monitoring.

2. Name two AWS services used for monitoring.

AWS CloudWatch and AWS CloudTrail

3. What does AWS CloudTrail track?

CloudTrail logs API calls made by or on behalf of an AWS account.

4. What is a log file?

A file that records events, activities, and messages generated by applications or systems.

5.Define AWS CloudWatch.

A monitoring service for AWS resources and applications that collects metrics and logs, and generates alarms.

6. Name one AWS seWhat is the purpose of logging in cloud environments?

To record activities and events for auditing, debugging, compliance, and security monitoring.

7. What is the function of AWS Security Hub?

It provides a comprehensive view of security alerts and compliance status across AWS accounts.

8. What is the use of VPC Flow Logs?

To capture information about IP traffic going to and from network interfaces in a VPC.

9. Give one method of collecting logs in AWS.

Using CloudWatch Agent to push system and application logs.

10. What does AWS GuardDuty do?

It is a threat detection service that continuously monitors for malicious activity and unauthorized behavior.

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11. Part B,C Questions

1. Explain how AWS CloudTrail and CloudWatch work together to provide comprehensive monitoring and logging.
2. Describe the process of capturing and analyzing log files in a cloud environment. Mention tools involved.
3. Discuss the advantages of using AWS Security Hub for centralized monitoring of security events.

PART-C (15 MARKS)

1. Design a complete logging and monitoring strategy for a web application hosted on AWS. Include log types, monitoring tools, and alerting mechanisms.
2. Compare AWS CloudTrail, CloudWatch Logs, and VPC Flow Logs in terms of purpose, data captured, and use cases. Include a table if needed.
3. Demonstrate with steps how to set up AWS Security Hub across multiple accounts and integrate findings from services like GuardDuty and Macie.

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12. GATE Questions

1. Which AWS service is primarily used for recording API activity in an AWS account?
   a) Amazon EC2
   b) AWS CloudTrail
   c) AWS Lambda
   d) Amazon Route 53
2. Which AWS service is mainly used for resource monitoring and alert generation?
   a) AWS IAM
   b) AWS Shield
   c) Amazon CloudWatch
   d) AWS Config
3. Which of the following is an example of a monitoring metric in CloudWatch?
   a) CPU Utilization
   b) IAM Policy
   c) S3 Bucket Name
   d) Security Group ID
4. Which AWS service provides centralized security findings and compliance status?
   a) AWS Security Hub
   b) AWS Lambda
   c) AWS Config
   d) Amazon Inspector
5. Which feature of CloudWatch triggers notifications based on thresholds?
   a) CloudWatch Alarm
   b) Event History
   c) VPC Flow Log
   d) Route Table

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12. GATE Questions

6. Which AWS service helps in collecting logs for VPC network interfaces?
a) AWS Shield
b) AWS WAF
c) VPC Flow Logs
d) Amazon Inspector
7. Which of the following is mainly used for auditing AWS account activities?
a) AWS CloudTrail
b) Amazon SNS
c) AWS Lambda
d) Amazon RDS
8. Which AWS service can visualize monitoring data using dashboards?
a) AWS IAM
b) Amazon CloudWatch
c) AWS Artifact
d) Amazon S3
9.The primary purpose of logging in cloud systems is to:
a) Increase hardware speed
b) Store application source code
c) Track activities and detect issues
d) Manage DNS records
10. Which AWS service integrates findings from GuardDuty, Inspector, and Macie?
a) AWS Config
b) AWS Security Hub
c) Amazon CloudFront
d) AWS Trusted Advisor

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13. Assessment Schedule

Assessment Tool	Proposed Date	Actual Date	Course Outcome	Program Outcome (Filled Gap)
Assessment I			CO1, CO2	
Assessment II			CO3, CO4	
Model			CO1, CO2, CO3, CO4, CO5,CO6	

14. Prescribed Text Books & Reference Books

TEXT BOOKS:

1. Tim Mather, Subra Kumaraswamy, Shahed Latif, "Cloud Security and Privacy, An Enterprise Perspective on Risks and Compliance", Oreilly Media 2009.

2. Vic (J.R.) Winkler, "Securing the Cloud, Cloud Computer Security Techniques and Tactics", Syngress, April 2011.

REFERENCES:

1. Rajkumar Buyya, James Broberg, Andrzej, —Cloud Computing: Principles and Paradigms‖, Wiley India Publications 2011.

2. Arshdeep Bahga and Vijay Madisetti, —Cloud Computing –A Hands on Approach‖, Universities Press (India) Pvt Ltd. 2014

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15. Mini Project suggestions

1. Real-Time Monitoring using Amazon CloudWatch

Monitor EC2 performance metrics.

Configure alarms and notifications.

Generate monitoring dashboards.

2. Centralized Logging using AWS CloudTrail

Capture AWS API activities.

Store logs in Amazon S3.

Analyze suspicious activities.

3. Security Dashboard using AWS Security Hub

Integrate GuardDuty and Inspector.

Display centralized security findings.

Generate compliance reports.

4. Log Analysis System using CloudWatch Logs

Collect logs from EC2 instances.

Detect unusual login attempts.

Generate alerts automatically.

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Thank you

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